



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

10 / 10 completed

Integer Array

INTEGER ARRAY

"Write a C program to demonstrate pointer arithmetic on an integer array. Display element values using both array indexing and pointer notation. Show address difference between consecutive elements.

Sample Input:

Enter number of elements: 4

Enter elements:

10 20 30 40

Sample Output:

Using Array Indexing:

arr[0] = 10

arr[1] = 20

arr[2] = 30

arr[3] = 40

Using Pointer Notation:

*(ptr+0) = 10

*(ptr+1) = 20

*(ptr+2) = 30

*(ptr+3) = 40

PROGRAM EDITOR

C

Run

Save

```
1
2 #include <stdio.h>
3
4 int main() {
5     int n, i;
6
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    int arr[n];
11    int *ptr = arr;
12
13    printf("Enter elements:\n");
14    for (i = 0; i < n; i++) {
15        scanf("%d", &arr[i]);
16    }
17
18
19    printf("\nUsing Array Indexing:\n");
20    for (i = 0; i < n; i++) {
21        printf("arr[%d] = %d\n", i, arr[i]):
```

OUTPUT

Enter number of elements: Enter elements:

Using Array Indexing:

arr[0] = 20

arr[1] = 30

arr[2] = 1771640409

arr[3] = 32758

arr[4] = 1771745307

arr[5] = 32758

arr[6] = -1673528612

INPUT

10 20 30

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